Classe cancel·lada setmana 14–18/10/2019 (grups 11 i/o 23)

21. *El problema de l'eliminació de superfícies ocultes és un problema important en informàtica gràfica. Si des de la teva perspectiva, en Pepet està davant d'en Ramonet, podràs veure en Pepet però no en Ramonet. Considereu el següent problema, restringit al pla. Us donen n rectes no verticals al pla, $L_1, \ldots, L_n$, on la recta L_i ve especificada per l'equació $y = a_i x + b_i$. Assumim, que no hi han tres rectes que es creuen exactament al mateix punt. Direm que L_i és maximal en x_0 de la coordenada x, si per qualsevol $1 \leq j \leq n$ amb $j \neq i$ tenim que $a_i x_0 + b_i > a_j x_0 + b_j$. Direm que L_i és* visible *si té algun punt maximal.*

    Donat com a entrada un conjunt de n rectes $\mathcal{L} = \{L_1, \ldots, L_n\}$, doneu un algorisme que, en temps $O(n \lg n)$, retorni les rectes no visibles.

    Solució:

    We first label the lines in order of increasing slope, and then use a divide-and-conquer approach. If $n \leq 3$ — the base case of the divide-and-conquer approach — we can easily find the visible lines in constant time. (The first and third lines will always be visible; the second will be visible if and only if it meets the first line to the left of where the third line meets the first line.)

    Let $m = \lceil n/2 \rceil$. We first recursively compute the sequence of visible lines among $L_1, \ldots, L_m$ — say they are $\mathcal{L} = \{L_{i_1}, \ldots, L_{i_p}\}$ in order of increasing slope. We also compute, in this recursive call, the sequence of points $a_1, \ldots, a_{p-1}$ where a_k is the intersection of line L_{i_k} with line $L_{i_{k+1}}$. Notice that $a_1, \ldots, a_{p-1}$ will have increasing x-coordinates; for if two lines are both visible, the region in which the line of smaller slope is uppermost lies to the left of the region in which the line of larger slope is uppermost. Similarly, we recursively compute the sequence $\mathcal{L}' = \{L_{j_1}, \ldots, L_{j_q}\}$ of visible lines among $L_{m+1}, \ldots, L_n$, together with the sequence of intersection points $b_k = L_{j_k} \cap L_{j_{k+1}}$ for $k = 1, \ldots, q-1$.

    To complete the algorithm, we must show how to determine the visible lines in $\mathcal{L} \cup \mathcal{L}'$, together with the corresponding intersection points, in $O(n)$ time. (Note that $p + q \leq n$, so it is enough to run in time $O(p+q)$.) We know that L_{i_1} will be visible, because it has the minimum slope among all the lines in this list; similarly, we know that L_{j_q} will be visible, because it has the maximum slope.

    We merge the sorted lists $a_1, \ldots, a_{p-1}$ and $b_1, \ldots, b_{q-1}$ into a single list of points $c_1, c_2, \ldots, c_{p+q-2}$ ordered by increasing x-coordinate. This takes $O(n)$ time. Now, for each k, we consider the line that is uppermost in $\mathcal{L}$ at x-coordinate c_k, and the line that is uppermost in $\mathcal{L}'$ at x-coordinate c_k. Let ℓ be the smallest index for which the uppermost line in $\mathcal{L}'$ lies above the uppermost line in $\mathcal{L}$ at x-coordinate c_ℓ. Let the two lines at this point be $L_{i_s} \in \mathcal{L}$ and $L_{j_t} \in \mathcal{L}'$. Let (x^*, y^*) denote the point in the plane at which L_{i_s} and L_{j_t} intersect. We have thus established that x^* lies between the x-coordinates of $c_{\ell-1}$ and c_ℓ. This means that L_{i_s} is uppermost in $\mathcal{L} \cup \mathcal{L}'$ immediately to the left of x^*, and L_{j_t} is uppermost in $\mathcal{L} \cup \mathcal{L}'$ immediately to the right of x^*. Consequently, the sequence of visible lines among $\mathcal{L} \cup \mathcal{L}'$ is $L_{i_1}, \ldots, L_{i_s}, L_{j_t}, \ldots, L_{j_q}$; and the sequence of intersection points is $a_{i_1}, \ldots, a_{i_{s-1}}, (x^*, y^*), b_{j_t}, \ldots, b_{j_{q-1}}$. Since this is what we need to return to the next level of the recursion, the algorithm is complete.

28. *Sigui S una estructura de dades que es una pila (stack), amb tres operacions: Les dues genèriques* **Push** (S,x)*, que col·loca x al cim de la pila i* **Pop** (S) *que retorna l'objecte x que es al cim de la pila. Considerem una nova operació* **Multipop** (S,k)*, que donat una pila amb mes de* $k \geq 0$ *elements retorna els k primers elements. Per ex. Si* $S = [23,17,6,39,10,47]$ *aleshores* **Multipop** $(S,4)$*, retornara seqüencialment els elements 23,17,6 i 39, i deixara la pila* $[10,47]$*. Si ara tornéssim a fer* **Multipop** $(S,4)$*. Retornaria 10 i 47 i deixaria buida la pila. Notem que el cost cas pitjor de* **Multipop** (S,k) *on S conte n elements, es* $= \min\{n,k\}$*.*
Si considerem una seqüencia de n operacions **Push, Pup, Multipop** *sobre una pila que inicialment es buida, el* **cas pitjo***r de* **Multipop** *pot ser* $O(n)$ *(per ex. fem* $n/2$ **Push** *a S i després fem* **Multipop** $(S,n/2)$*. Per tant, el cas pitjor de fer* **Multipop** *sobre n elements en* $O(n^2)$*.*
Demostreu que utilitzant anàlisi amortitzant podem obtenir un millor cost amortitzat, fent servir:

   (a) el mètode d'agregació.

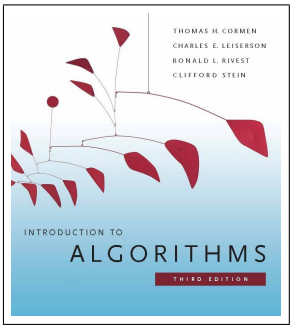
   (b) el mètode de l'estalviador (accounting).

Solució:

Podeu trobar la solució a aquest exercici com a exemple d'aplicació dels dos mètodes (agregació i estalviador) a [1], Capítol 17: "Amortized Analysis". S'adjunta una còpia com a annex a aquest document.

Referències

[1] Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein. *Introduction to Algorithms, 3rd Edition*. MIT Press, 2009.



29. *Donat un multiconjunt dinàmic d'enters, S, que pot tenir valors duplicats, dissenyeu una estructura de dades que done suport a les següents operacions: INSERT(S,x), insereix un enter x a S, i DELETE-LARGER-HALF(S), elimina els $\lfloor |S|/2 \rfloor$ elements més grans de S.*
Expliqueu com implementaríeu aquesta estructura de dades de manera que qualsevol seqüència de m operacions INSERT i DELETE-LARGER-HALF, es pugui fer en $O(m)$ passos. La vostra implemenatació també ha d'incloure una manera de treure els elements de S en $O(|S|)$ passos.

Solució:

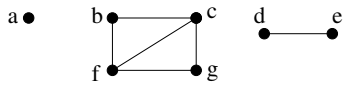
Una possible solució és un algorisme que insereixi (pel final) els elements a un vector quan es fa un INSERT i, quan arribava a un DELETE-LARGER-HALF, calculi la mediana (per selecció determinista) i copiï els $n/2$ elements menors a un nou vector. Per a veure que el cost amortitzat per element és constant utilitzarem *accounting*:

- Per a cada inserció carreguem 6 unitats i per a cada esborrat 0.
- Demostrem que, en qualsevol moment donat, si el vector té k elements, aleshores tindrem crèdit $5k$:
  - En inserir el primer element, gastem de veritat 1 unitat i em queden 5 unitats de crèdit.
  - Suposant que tinc crèdit $5k$, si afegim un element passarem a tenir $5k+5$ i $k+1$ elements.
  - Si l'operació és un esborrat, passaré de k a $k/2$ elements al vector. Gastarem: k unitats en calcular la mediana, k unitats en recórrer i comparar el vector, i $k/2$ en copiar la meitat dels elements. Per tant, acabem tenint un crèdit de:

    $$5k - 2k - \frac{k}{2} = \frac{10-4-1}{2}\,k = 5\,\frac{k}{2}.$$

    Per tant, el cost per element és $O(1)$ i, el total és, doncs, $O(m)$.

30. *Donat un graf no dirigit $G = (V,E)$ amb n vèrtexs i m arestes, dissenyeu un algorisme que utilitzant UNION-FIND (tal com s'ha explicat a classe) ens digui si G és o no connex i en cas de que no ho sigui, puguem averiguar a quina component connexa pertany un vèrtex v donat. Quina és la complexitat del vostre algorisme? Apliqueu el vostre algorisme al següent exemple:*



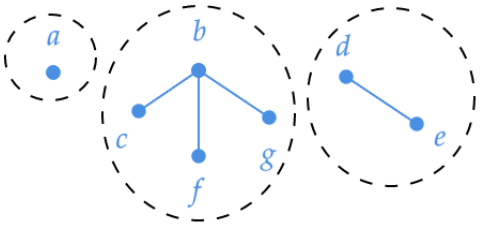
El problema de trobar la connectivitat dún graf no dirigit té complexitat $O(n+m)$ utilitzant DFS o BFS, en quins casos UNION-FIND és més ràpid?

Solució:

Posem pes 1 a totes les arestes del graf. Cada vèrtex és el seu propi conjunt. A continuació fem servir l'algorisme de Kruskal fins que no poguem aplicar més UNION. Per a saber la component connexa d'un vèrtex u faríem FIND(u).

El cost temporal del DFS/BFS és $O(n+m)$, mentre que pel UNION-FIND amb compressió de camins és $O(m\,\alpha(n))$, on $\alpha(n)$ és la funció inversa d'Ackermann[1].

Aplicat sobre el graf de l'exemple obtindríem:



L'algorisme UNION-FIND s'escau millor per a situacions on les relacions d'equivalència són canviants. Així, quan les arestes del graf són dinàmiques (canvien en el temps), els recorreguts DFS/BFS no són una bona elecció perquè no es poden aplicar progressivament, mentre que el UNION-FIND és una bona opció perquè pot adaptar fàcilment la llista de component connexes als canvis que s'hi succeeixen.

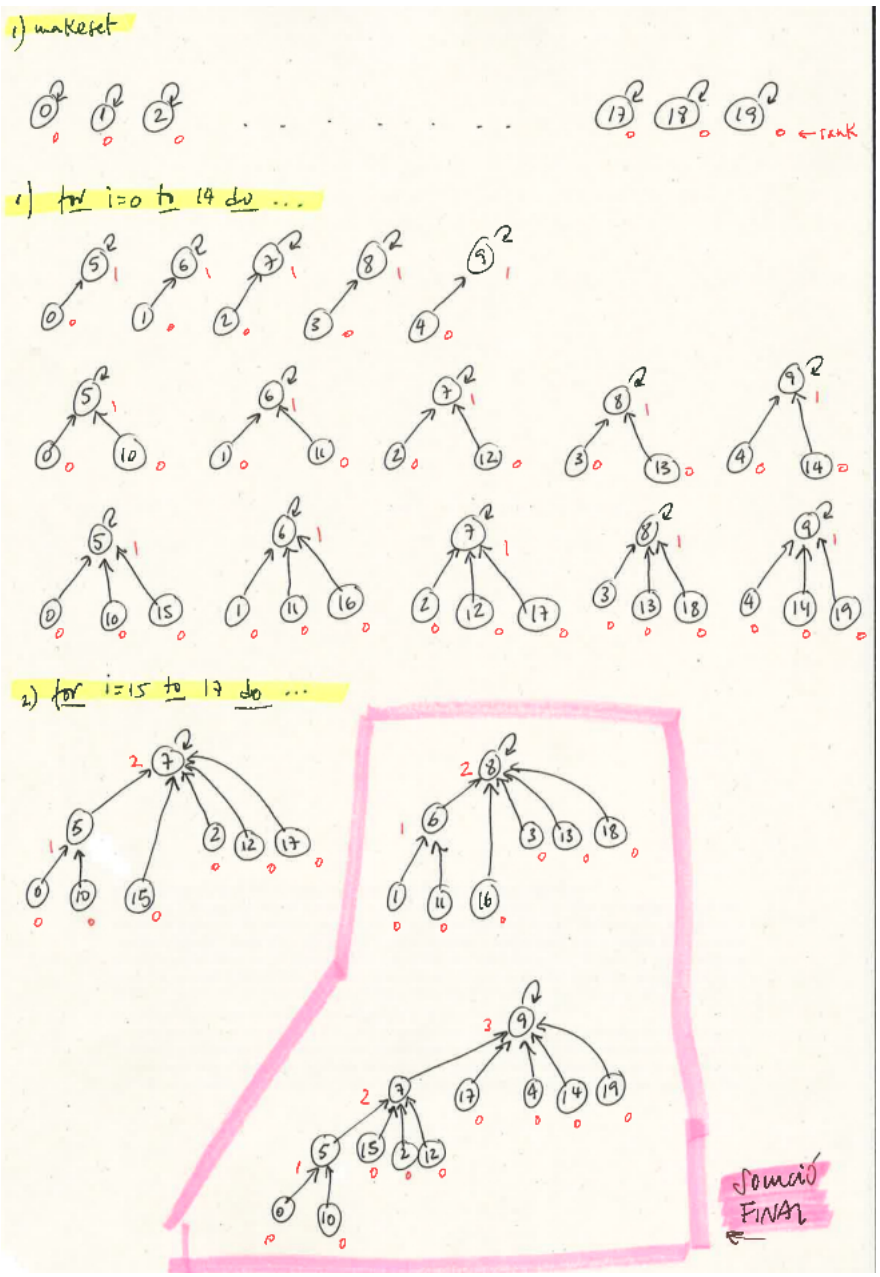
[1] Donat que la funció d'Ackermann $f(n) = A(n,n)$ creix molt ràpid, la seva funció inversa $f^{-1}(n)$ ho fa molt lentament. Aquesta funció inversa es denota normalment amb $\alpha(n)$. De fet, $\alpha(n)$ és menys de 5 per a qualsevol tamany d'entrada n pràctic.

31. *Tenim 20 elements i fem un* MakeSet*. Suposem que els conjunts resultants estan enumerats de 0 fins a 19. Executem el següent codi:*

for $i = 0$ **to** 14 **do**
  union(**find**(i), **find**$(i+5)$)
end for
for $i = 15$ **to** 17 **do**
  union(**find**(i), **find**$(i+2)$)
end for

Dibuixeu la seqüencia d'arbres fins a arribar al resultat final (dos arbres) quan la implementació utilitza unió per rang amb compressió de camins. Doneu el rang dels nusos al dos arbres resultants.

Solució:



Annex

17 Amortized Analysis

In an ***amortized analysis***, we average the time required to perform a sequence of data-structure operations over all the operations performed. With amortized analysis, we can show that the average cost of an operation is small, if we average over a sequence of operations, even though a single operation within the sequence might be expensive. Amortized analysis differs from average-case analysis in that probability is not involved; an amortized analysis guarantees the *average performance of each operation in the worst case.*

The first three sections of this chapter cover the three most common techniques used in amortized analysis. Section 17.1 starts with aggregate analysis, in which we determine an upper bound $T(n)$ on the total cost of a sequence of n operations. The average cost per operation is then $T(n)/n$. We take the average cost as the amortized cost of each operation, so that all operations have the same amortized cost.

Section 17.2 covers the accounting method, in which we determine an amortized cost of each operation. When there is more than one type of operation, each type of operation may have a different amortized cost. The accounting method overcharges some operations early in the sequence, storing the overcharge as "prepaid credit" on specific objects in the data structure. Later in the sequence, the credit pays for operations that are charged less than they actually cost.

Section 17.3 discusses the potential method, which is like the accounting method in that we determine the amortized cost of each operation and may overcharge operations early on to compensate for undercharges later. The potential method maintains the credit as the "potential energy" of the data structure as a whole instead of associating the credit with individual objects within the data structure.

We shall use two examples to examine these three methods. One is a stack with the additional operation MULTIPOP, which pops several objects at once. The other is a binary counter that counts up from 0 by means of the single operation INCREMENT.

While reading this chapter, bear in mind that the charges assigned during an amortized analysis are for analysis purposes only. They need not—and should not—appear in the code. If, for example, we assign a credit to an object x when using the accounting method, we have no need to assign an appropriate amount to some attribute, such as $x.credit$, in the code.

When we perform an amortized analysis, we often gain insight into a particular data structure, and this insight can help us optimize the design. In Section 17.4, for example, we shall use the potential method to analyze a dynamically expanding and contracting table.

17.1 Aggregate analysis

In ***aggregate analysis***, we show that for all n, a sequence of n operations takes *worst-case* time $T(n)$ in total. In the worst case, the average cost, or ***amortized cost***, per operation is therefore $T(n)/n$. Note that this amortized cost applies to each operation, even when there are several types of operations in the sequence. The other two methods we shall study in this chapter, the accounting method and the potential method, may assign different amortized costs to different types of operations.

Stack operations

In our first example of aggregate analysis, we analyze stacks that have been augmented with a new operation. Section 10.1 presented the two fundamental stack operations, each of which takes $O(1)$ time:

PUSH(S, x) pushes object x onto stack S.

POP(S) pops the top of stack S and returns the popped object. Calling POP on an empty stack generates an error.

Since each of these operations runs in $O(1)$ time, let us consider the cost of each to be 1. The total cost of a sequence of n PUSH and POP operations is therefore n, and the actual running time for n operations is therefore $\Theta(n)$.

Now we add the stack operation MULTIPOP(S, k), which removes the k top objects of stack S, popping the entire stack if the stack contains fewer than k objects. Of course, we assume that k is positive; otherwise the MULTIPOP operation leaves the stack unchanged. In the following pseudocode, the operation STACK-EMPTY returns TRUE if there are no objects currently on the stack, and FALSE otherwise.

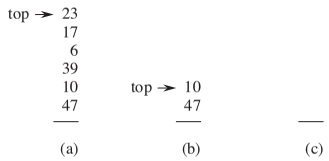


Figure 17.1 The action of MULTIPOP on a stack S, shown initially in **(a)**. The top 4 objects are popped by MULTIPOP$(S, 4)$, whose result is shown in **(b)**. The next operation is MULTIPOP$(S, 7)$, which empties the stack—shown in **(c)**—since there were fewer than 7 objects remaining.

```
MULTIPOP(S, k)
1  while not STACK-EMPTY(S) and k > 0
2      POP(S)
3      k = k - 1
```

Figure 17.1 shows an example of MULTIPOP.

What is the running time of MULTIPOP(S, k) on a stack of s objects? The actual running time is linear in the number of POP operations actually executed, and thus we can analyze MULTIPOP in terms of the abstract costs of 1 each for PUSH and POP. The number of iterations of the **while** loop is the number $\min(s, k)$ of objects popped off the stack. Each iteration of the loop makes one call to POP in line 2. Thus, the total cost of MULTIPOP is $\min(s, k)$, and the actual running time is a linear function of this cost.

Let us analyze a sequence of n PUSH, POP, and MULTIPOP operations on an initially empty stack. The worst-case cost of a MULTIPOP operation in the sequence is $O(n)$, since the stack size is at most n. The worst-case time of any stack operation is therefore $O(n)$, and hence a sequence of n operations costs $O(n^2)$, since we may have $O(n)$ MULTIPOP operations costing $O(n)$ each. Although this analysis is correct, the $O(n^2)$ result, which we obtained by considering the worst-case cost of each operation individually, is not tight.

Using aggregate analysis, we can obtain a better upper bound that considers the entire sequence of n operations. In fact, although a single MULTIPOP operation can be expensive, any sequence of n PUSH, POP, and MULTIPOP operations on an initially empty stack can cost at most $O(n)$. Why? We can pop each object from the stack at most once for each time we have pushed it onto the stack. Therefore, the number of times that POP can be called on a nonempty stack, including calls within MULTIPOP, is at most the number of PUSH operations, which is at most n. For any value of n, any sequence of n PUSH, POP, and MULTIPOP operations takes a total of $O(n)$ time. The average cost of an operation is $O(n)/n = O(1)$. In aggregate

analysis, we assign the amortized cost of each operation to be the average cost. In this example, therefore, all three stack operations have an amortized cost of $O(1)$.

We emphasize again that although we have just shown that the average cost, and hence the running time, of a stack operation is $O(1)$, we did not use probabilistic reasoning. We actually showed a *worst-case* bound of $O(n)$ on a sequence of n operations. Dividing this total cost by n yielded the average cost per operation, or the amortized cost.

Incrementing a binary counter

As another example of aggregate analysis, consider the problem of implementing a k-bit binary counter that counts upward from 0. We use an array $A[0\,..\,k-1]$ of bits, where $A.length = k$, as the counter. A binary number x that is stored in the counter has its lowest-order bit in $A[0]$ and its highest-order bit in $A[k-1]$, so that $x = \sum_{i=0}^{k-1} A[i] \cdot 2^i$. Initially, $x = 0$, and thus $A[i] = 0$ for $i = 0, 1, \ldots, k-1$. To add 1 (modulo 2^k) to the value in the counter, we use the following procedure.

```
INCREMENT(A)
1  i = 0
2  while i < A.length and A[i] == 1
3      A[i] = 0
4      i = i + 1
5  if i < A.length
6      A[i] = 1
```

Figure 17.2 shows what happens to a binary counter as we increment it 16 times, starting with the initial value 0 and ending with the value 16. At the start of each iteration of the **while** loop in lines 2–4, we wish to add a 1 into position i. If $A[i] = 1$, then adding 1 flips the bit to 0 in position i and yields a carry of 1, to be added into position $i + 1$ on the next iteration of the loop. Otherwise, the loop ends, and then, if $i < k$, we know that $A[i] = 0$, so that line 6 adds a 1 into position i, flipping the 0 to a 1. The cost of each INCREMENT operation is linear in the number of bits flipped.

As with the stack example, a cursory analysis yields a bound that is correct but not tight. A single execution of INCREMENT takes time $\Theta(k)$ in the worst case, in which array A contains all 1s. Thus, a sequence of n INCREMENT operations on an initially zero counter takes time $O(nk)$ in the worst case.

We can tighten our analysis to yield a worst-case cost of $O(n)$ for a sequence of n INCREMENT operations by observing that not all bits flip each time INCREMENT is called. As Figure 17.2 shows, $A[0]$ does flip each time INCREMENT is called. The next bit up, $A[1]$, flips only every other time: a sequence of n INCREMENT

Counter value	A[7]	A[6]	A[5]	A[4]	A[3]	A[2]	A[1]	A[0]	Total cost
0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	1	1
2	0	0	0	0	0	0	1	0	3
3	0	0	0	0	0	0	1	1	4
4	0	0	0	0	0	1	0	0	7
5	0	0	0	0	0	1	0	1	8
6	0	0	0	0	0	1	1	0	10
7	0	0	0	0	0	1	1	1	11
8	0	0	0	0	1	0	0	0	15
9	0	0	0	0	1	0	0	1	16
10	0	0	0	0	1	0	1	0	18
11	0	0	0	0	1	0	1	1	19
12	0	0	0	0	1	1	0	0	22
13	0	0	0	0	1	1	0	1	23
14	0	0	0	0	1	1	1	0	25
15	0	0	0	0	1	1	1	1	26
16	0	0	0	1	0	0	0	0	31

Figure 17.2 An 8-bit binary counter as its value goes from 0 to 16 by a sequence of 16 INCREMENT operations. Bits that flip to achieve the next value are shaded. The running cost for flipping bits is shown at the right. Notice that the total cost is always less than twice the total number of INCREMENT operations.

operations on an initially zero counter causes $A[1]$ to flip $\lfloor n/2 \rfloor$ times. Similarly, bit $A[2]$ flips only every fourth time, or $\lfloor n/4 \rfloor$ times in a sequence of n INCREMENT operations. In general, for $i = 0, 1, \ldots, k-1$, bit $A[i]$ flips $\lfloor n/2^i \rfloor$ times in a sequence of n INCREMENT operations on an initially zero counter. For $i \geq k$, bit $A[i]$ does not exist, and so it cannot flip. The total number of flips in the sequence is thus

$$\begin{aligned} \sum_{i=0}^{k-1} \left\lfloor \frac{n}{2^i} \right\rfloor &< n \sum_{i=0}^{\infty} \frac{1}{2^i} \\ &= 2n , \end{aligned}$$

by equation (A.6). The worst-case time for a sequence of n INCREMENT operations on an initially zero counter is therefore $O(n)$. The average cost of each operation, and therefore the amortized cost per operation, is $O(n)/n = O(1)$.

Exercises

17.1-1

If the set of stack operations included a MULTIPUSH operation, which pushes k items onto the stack, would the $O(1)$ bound on the amortized cost of stack operations continue to hold?

17.1-2

Show that if a DECREMENT operation were included in the k-bit counter example, n operations could cost as much as $\Theta(nk)$ time.

17.1-3

Suppose we perform a sequence of n operations on a data structure in which the ith operation costs i if i is an exact power of 2, and 1 otherwise. Use aggregate analysis to determine the amortized cost per operation.

17.2 The accounting method

In the ***accounting method*** of amortized analysis, we assign differing charges to different operations, with some operations charged more or less than they actually cost. We call the amount we charge an operation its ***amortized cost***. When an operation's amortized cost exceeds its actual cost, we assign the difference to specific objects in the data structure as ***credit***. Credit can help pay for later operations whose amortized cost is less than their actual cost. Thus, we can view the amortized cost of an operation as being split between its actual cost and credit that is either deposited or used up. Different operations may have different amortized costs. This method differs from aggregate analysis, in which all operations have the same amortized cost.

We must choose the amortized costs of operations carefully. If we want to show that in the worst case the average cost per operation is small by analyzing with amortized costs, we must ensure that the total amortized cost of a sequence of operations provides an upper bound on the total actual cost of the sequence. Moreover, as in aggregate analysis, this relationship must hold for all sequences of operations. If we denote the actual cost of the ith operation by c_i and the amortized cost of the ith operation by $\widehat{c}_i$, we require

$$\sum_{i=1}^{n} \widehat{c}_i \geq \sum_{i=1}^{n} c_i \tag{17.1}$$

for all sequences of n operations. The total credit stored in the data structure is the difference between the total amortized cost and the total actual cost, or

$\sum_{i=1}^{n} \hat{c}_i - \sum_{i=1}^{n} c_i$. By inequality (17.1), the total credit associated with the data structure must be nonnegative at all times. If we ever were to allow the total credit to become negative (the result of undercharging early operations with the promise of repaying the account later on), then the total amortized costs incurred at that time would be below the total actual costs incurred; for the sequence of operations up to that time, the total amortized cost would not be an upper bound on the total actual cost. Thus, we must take care that the total credit in the data structure never becomes negative.

Stack operations

To illustrate the accounting method of amortized analysis, let us return to the stack example. Recall that the actual costs of the operations were

PUSH	1 ,
POP	1 ,
MULTIPOP	$\min(k, s)$,

where k is the argument supplied to MULTIPOP and s is the stack size when it is called. Let us assign the following amortized costs:

PUSH	2 ,
POP	0 ,
MULTIPOP	0 .

Note that the amortized cost of MULTIPOP is a constant (0), whereas the actual cost is variable. Here, all three amortized costs are constant. In general, the amortized costs of the operations under consideration may differ from each other, and they may even differ asymptotically.

We shall now show that we can pay for any sequence of stack operations by charging the amortized costs. Suppose we use a dollar bill to represent each unit of cost. We start with an empty stack. Recall the analogy of Section 10.1 between the stack data structure and a stack of plates in a cafeteria. When we push a plate on the stack, we use 1 dollar to pay the actual cost of the push and are left with a credit of 1 dollar (out of the 2 dollars charged), which we leave on top of the plate. At any point in time, every plate on the stack has a dollar of credit on it.

The dollar stored on the plate serves as prepayment for the cost of popping it from the stack. When we execute a POP operation, we charge the operation nothing and pay its actual cost using the credit stored in the stack. To pop a plate, we take the dollar of credit off the plate and use it to pay the actual cost of the operation. Thus, by charging the PUSH operation a little bit more, we can charge the POP operation nothing.

Moreover, we can also charge MULTIPOP operations nothing. To pop the first plate, we take the dollar of credit off the plate and use it to pay the actual cost of a POP operation. To pop a second plate, we again have a dollar of credit on the plate to pay for the POP operation, and so on. Thus, we have always charged enough up front to pay for MULTIPOP operations. In other words, since each plate on the stack has 1 dollar of credit on it, and the stack always has a nonnegative number of plates, we have ensured that the amount of credit is always nonnegative. Thus, for *any* sequence of n PUSH, POP, and MULTIPOP operations, the total amortized cost is an upper bound on the total actual cost. Since the total amortized cost is $O(n)$, so is the total actual cost.

Incrementing a binary counter

As another illustration of the accounting method, we analyze the INCREMENT operation on a binary counter that starts at zero. As we observed earlier, the running time of this operation is proportional to the number of bits flipped, which we shall use as our cost for this example. Let us once again use a dollar bill to represent each unit of cost (the flipping of a bit in this example).

For the amortized analysis, let us charge an amortized cost of 2 dollars to set a bit to 1. When a bit is set, we use 1 dollar (out of the 2 dollars charged) to pay for the actual setting of the bit, and we place the other dollar on the bit as credit to be used later when we flip the bit back to 0. At any point in time, every 1 in the counter has a dollar of credit on it, and thus we can charge nothing to reset a bit to 0; we just pay for the reset with the dollar bill on the bit.

Now we can determine the amortized cost of INCREMENT. The cost of resetting the bits within the **while** loop is paid for by the dollars on the bits that are reset. The INCREMENT procedure sets at most one bit, in line 6, and therefore the amortized cost of an INCREMENT operation is at most 2 dollars. The number of 1s in the counter never becomes negative, and thus the amount of credit stays nonnegative at all times. Thus, for n INCREMENT operations, the total amortized cost is $O(n)$, which bounds the total actual cost.

Exercises

17.2-1

Suppose we perform a sequence of stack operations on a stack whose size never exceeds k. After every k operations, we make a copy of the entire stack for backup purposes. Show that the cost of n stack operations, including copying the stack, is $O(n)$ by assigning suitable amortized costs to the various stack operations.

17.2-2
Redo Exercise 17.1-3 using an accounting method of analysis.

17.2-3
Suppose we wish not only to increment a counter but also to reset it to zero (i.e., make all bits in it 0). Counting the time to examine or modify a bit as $\Theta(1)$, show how to implement a counter as an array of bits so that any sequence of n INCREMENT and RESET operations takes time $O(n)$ on an initially zero counter. (*Hint:* Keep a pointer to the high-order 1.)

17.3 The potential method

Instead of representing prepaid work as credit stored with specific objects in the data structure, the ***potential method*** of amortized analysis represents the prepaid work as "potential energy," or just "potential," which can be released to pay for future operations. We associate the potential with the data structure as a whole rather than with specific objects within the data structure.

The potential method works as follows. We will perform n operations, starting with an initial data structure D_0. For each $i = 1, 2, \ldots, n$, we let c_i be the actual cost of the ith operation and D_i be the data structure that results after applying the ith operation to data structure D_{i-1}. A ***potential function*** Φ maps each data structure D_i to a real number $\Phi(D_i)$, which is the ***potential*** associated with data structure D_i. The ***amortized cost*** $\hat{c}_i$ of the ith operation with respect to potential function Φ is defined by

$$\hat{c}_i = c_i + \Phi(D_i) - \Phi(D_{i-1}) . \tag{17.2}$$

The amortized cost of each operation is therefore its actual cost plus the change in potential due to the operation. By equation (17.2), the total amortized cost of the n operations is

$$\begin{aligned} \sum_{i=1}^{n} \hat{c}_i &= \sum_{i=1}^{n} (c_i + \Phi(D_i) - \Phi(D_{i-1})) \\ &= \sum_{i=1}^{n} c_i + \Phi(D_n) - \Phi(D_0) . \end{aligned} \tag{17.3}$$

The second equality follows from equation (A.9) because the $\Phi(D_i)$ terms telescope.

If we can define a potential function Φ so that $\Phi(D_n) \geq \Phi(D_0)$, then the total amortized cost $\sum_{i=1}^{n} \hat{c}_i$ gives an upper bound on the total actual cost $\sum_{i=1}^{n} c_i$.

In practice, we do not always know how many operations might be performed. Therefore, if we require that $\Phi(D_i) \geq \Phi(D_0)$ for all i, then we guarantee, as in the accounting method, that we pay in advance. We usually just define $\Phi(D_0)$ to be 0 and then show that $\Phi(D_i) \geq 0$ for all i. (See Exercise 17.3-1 for an easy way to handle cases in which $\Phi(D_0) \neq 0$.)

Intuitively, if the potential difference $\Phi(D_i) - \Phi(D_{i-1})$ of the ith operation is positive, then the amortized cost $\hat{c}_i$ represents an overcharge to the ith operation, and the potential of the data structure increases. If the potential difference is negative, then the amortized cost represents an undercharge to the ith operation, and the decrease in the potential pays for the actual cost of the operation.

The amortized costs defined by equations (17.2) and (17.3) depend on the choice of the potential function Φ. Different potential functions may yield different amortized costs yet still be upper bounds on the actual costs. We often find trade-offs that we can make in choosing a potential function; the best potential function to use depends on the desired time bounds.

Stack operations

To illustrate the potential method, we return once again to the example of the stack operations PUSH, POP, and MULTIPOP. We define the potential function Φ on a stack to be the number of objects in the stack. For the empty stack D_0 with which we start, we have $\Phi(D_0) = 0$. Since the number of objects in the stack is never negative, the stack D_i that results after the ith operation has nonnegative potential, and thus

$$\begin{aligned} \Phi(D_i) &\geq 0 \\ &= \Phi(D_0) . \end{aligned}$$

The total amortized cost of n operations with respect to Φ therefore represents an upper bound on the actual cost.

Let us now compute the amortized costs of the various stack operations. If the ith operation on a stack containing s objects is a PUSH operation, then the potential difference is

$$\begin{aligned} \Phi(D_i) - \Phi(D_{i-1}) &= (s+1) - s \\ &= 1 . \end{aligned}$$

By equation (17.2), the amortized cost of this PUSH operation is

$$\begin{aligned} \hat{c}_i &= c_i + \Phi(D_i) - \Phi(D_{i-1}) \\ &= 1 + 1 \\ &= 2 . \end{aligned}$$

Suppose that the ith operation on the stack is MULTIPOP(S, k), which causes $k' = \min(k, s)$ objects to be popped off the stack. The actual cost of the operation is k', and the potential difference is

$$\Phi(D_i) - \Phi(D_{i-1}) = -k' .$$

Thus, the amortized cost of the MULTIPOP operation is

$$\begin{aligned} \hat{c}_i &= c_i + \Phi(D_i) - \Phi(D_{i-1}) \\ &= k' - k' \\ &= 0 . \end{aligned}$$

Similarly, the amortized cost of an ordinary POP operation is 0.

The amortized cost of each of the three operations is $O(1)$, and thus the total amortized cost of a sequence of n operations is $O(n)$. Since we have already argued that $\Phi(D_i) \geq \Phi(D_0)$, the total amortized cost of n operations is an upper bound on the total actual cost. The worst-case cost of n operations is therefore $O(n)$.

Incrementing a binary counter

As another example of the potential method, we again look at incrementing a binary counter. This time, we define the potential of the counter after the ith INCREMENT operation to be b_i, the number of 1s in the counter after the ith operation.

Let us compute the amortized cost of an INCREMENT operation. Suppose that the ith INCREMENT operation resets t_i bits. The actual cost of the operation is therefore at most $t_i + 1$, since in addition to resetting t_i bits, it sets at most one bit to 1. If $b_i = 0$, then the ith operation resets all k bits, and so $b_{i-1} = t_i = k$. If $b_i > 0$, then $b_i = b_{i-1} - t_i + 1$. In either case, $b_i \leq b_{i-1} - t_i + 1$, and the potential difference is

$$\begin{aligned} \Phi(D_i) - \Phi(D_{i-1}) &\leq (b_{i-1} - t_i + 1) - b_{i-1} \\ &= 1 - t_i . \end{aligned}$$

The amortized cost is therefore

$$\begin{aligned} \hat{c}_i &= c_i + \Phi(D_i) - \Phi(D_{i-1}) \\ &\leq (t_i + 1) + (1 - t_i) \\ &= 2 . \end{aligned}$$

If the counter starts at zero, then $\Phi(D_0) = 0$. Since $\Phi(D_i) \geq 0$ for all i, the total amortized cost of a sequence of n INCREMENT operations is an upper bound on the total actual cost, and so the worst-case cost of n INCREMENT operations is $O(n)$.

The potential method gives us an easy way to analyze the counter even when it does not start at zero. The counter starts with b_0 1s, and after n INCREMENT

operations it has b_n 1s, where $0 \leq b_0, b_n \leq k$. (Recall that k is the number of bits in the counter.) We can rewrite equation (17.3) as

$$\sum_{i=1}^{n} c_i = \sum_{i=1}^{n} \hat{c}_i - \Phi(D_n) + \Phi(D_0) . \tag{17.4}$$

We have $\hat{c}_i \leq 2$ for all $1 \leq i \leq n$. Since $\Phi(D_0) = b_0$ and $\Phi(D_n) = b_n$, the total actual cost of n INCREMENT operations is

$$\begin{aligned} \sum_{i=1}^{n} c_i &\leq \sum_{i=1}^{n} 2 - b_n + b_0 \\ &= 2n - b_n + b_0 . \end{aligned}$$

Note in particular that since $b_0 \leq k$, as long as $k = O(n)$, the total actual cost is $O(n)$. In other words, if we execute at least $n = \Omega(k)$ INCREMENT operations, the total actual cost is $O(n)$, no matter what initial value the counter contains.

Exercises

17.3-1
Suppose we have a potential function Φ such that $\Phi(D_i) \geq \Phi(D_0)$ for all i, but $\Phi(D_0) \neq 0$. Show that there exists a potential function Φ' such that $\Phi'(D_0) = 0$, $\Phi'(D_i) \geq 0$ for all $i \geq 1$, and the amortized costs using Φ' are the same as the amortized costs using Φ.

17.3-2
Redo Exercise 17.1-3 using a potential method of analysis.

17.3-3
Consider an ordinary binary min-heap data structure with n elements supporting the instructions INSERT and EXTRACT-MIN in $O(\lg n)$ worst-case time. Give a potential function Φ such that the amortized cost of INSERT is $O(\lg n)$ and the amortized cost of EXTRACT-MIN is $O(1)$, and show that it works.

17.3-4
What is the total cost of executing n of the stack operations PUSH, POP, and MULTIPOP, assuming that the stack begins with s_0 objects and finishes with s_n objects?

17.3-5
Suppose that a counter begins at a number with b 1s in its binary representation, rather than at 0. Show that the cost of performing n INCREMENT operations is $O(n)$ if $n = \Omega(b)$. (Do not assume that b is constant.)